

AI feel you: customer experience assessment via chatbot interviews

CE assessment
via chatbot
interviews

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Abstract

Purpose – While customer experience (CE) is recognized as a critical determinant of business success, both academics and managers are yet to find a means to gain a comprehensive understanding of CE cost-effectively. The authors argue that the application of relevant AI technology could help address this challenge. Employing interactively prompted narrative storytelling, and the authors investigate the effectiveness of sentiment analysis (SA) on extracting valuable CE insights from primary qualitative data generated via chatbot interviews.

Design/methodology/approach – Drawing on a granular and semantically clear framework for studying CE feelings, an artificial intelligence (AI) augmented chatbot was designed. The chatbot interviewed a crowdsourced sample of consumers about their recalled service experience feelings. By combining free-text and closed-ended questions, the authors were able to compare extracted sentiment polarities against established measurement scales and empirically validate our novel approach.

Findings – The authors demonstrate that SA can effectively extract CE feelings from primary chatbot data. This findings also suggest that further enhancement in accuracy can be achieved via improvements in the interplay between the chatbot interviewer and SA extraction algorithms.

Research limitations/implications – The proposed customer-centric approach can help service companies to study and better understand CE feelings in a cost-effective and scalable manner. The AI-augmented chatbots can also help companies to foster immersive and engaging relationships with customers. This study focuses on feelings, warranting further research on AI's value in studying other CE elements.

Originality/value – The unique inquisitive role of AI-infused chatbots in conducting interviews and analyzing data in realtime, offers considerable potential for studying CE and other subjective constructs.

Keywords Customer experience, Customer feelings, Sentiment analysis, Chatbot, Artificial intelligence, Storytelling

Paper type Research paper

Received 12 November 2019

Revised 18 April 2020

10 June 2020

12 June 2020

Accepted 21 June 2020

1. Introduction

Understanding customer experience (CE), comprised of experiential elements (e.g. cognitive and emotional), is paramount for service organizations aiming to successfully co-create value with their customers (McColl-Kennedy *et al.*, 2015). As a result, CE has remained an area of interest for both managers (McIntyre and Virzi, 2018) and researchers (Marketing Science Institute, 2018) for more than 50 years (Lemon and Verhoef, 2016).

Despite the growth of CE research, the body of literature remains fragmented and overlapping (Kranzbühler *et al.*, 2018), mainly focusing on the antecedents and outcomes of CE rather than defining and understanding it as a holistic, complex and subjective phenomenon (Kawaf and Tagg, 2017). Additionally, attempts to assess CE using survey approaches tend to possess an organizational bias. Such approaches focus on aspects deemed to be important by researchers or managers and not the elements of a context-specific experience, perceived significant by the customer (Ordeneš *et al.*, 2014). Specifically, they fail to recognize that CE is perceived and thus should also be measured within the domain of an individual's overall human experience (Fisk *et al.*, 2020). To address this, more appropriate approaches to study holistic CE, spanning the “digital, physical and social realms” of today's digital age, are required (Bolton *et al.*, 2018, p. 777; Holmlund *et al.*, 2020).

Arguably, phenomenological and introspective approaches – involving a process of narrative inquiry via data collection mechanisms such as storytelling and interviews



Journal of Service Management
Vol. 31 No. 4, 2020
pp. 745-766
© Emerald Publishing Limited
1757-5818
DOI 10.1108/JOSM-11-2019-0341

(Connelly and Clandinin, 1990) – are better suited for capturing comprehensive understanding of experiential constructs such as CE (Carù and Cova, 2003; Holbrook and Hirschman, 1982). This is because, in storytelling, key experiential elements like thoughts, feelings and behaviors are exchanged in the form of encompassing narratives, which are useful for studying critical events (Webster, 2007) and understanding CE in context.

While attempts have already been made to understand CE via storytelling in service research (e.g. Gentile *et al.*, 2007), these methods largely remain within the confines of interviews conducted by researchers. Such qualitative approaches constitute a minority of service research as they are more resource, time and cost-intensive compared to more efficient but potentially less informative quantitative methods (Benoit *et al.*, 2017). Finding an approach that is both highly informative and efficient has until now seemed unattainable. However, contemporary services firms could now attain the “best of both worlds”.

This could be achieved by employing chatbots and exploiting their ability to engage customers via storytelling. With growing recognition of their potential in service research (Kumar *et al.*, 2016), chatbots equipped with artificial intelligence (AI) could automatically extract CE from narrative conversations with customers – using a sentiment analysis (SA) algorithm – and hence contribute to CE theory. This could provide a useful starting point for a cost-effective service excellence strategy for service organizations (Ordenes and Zhang, 2019; Robinson *et al.*, 2019; Wirtz and Zeithaml, 2018).

The use of AI methods generally, SA in particular, calls for high semantic clarity (i.e. precise mapping to linguistic meanings). As a result, a bottom-up abductive approach was adopted to develop a granular and semantically clear model for feelings, one of the five elements of CE: thought, feeling, sensation, activity and relation (De Keyser *et al.*, 2020; Schmitt, 1999). There are several reasons why focusing on these elements individually and on feelings in particular, could yield advantageous results. First, there is an academic need to address the challenges in measuring feelings in CE (Richins, 1997) as well as understanding them better using empirically validated multi-methods (McColl-Kennedy *et al.*, 2015). Feelings relating to outcomes of service encounters are holistically connected with a person’s overall assessment of their human experience. Thus, an improved understanding of this CE element would aid managers to co-create service encounters with their customers to better satisfy their needs (Fisk *et al.*, 2020). Second, focusing on the CE feeling element addresses managerially-driven research priorities of (1) establishing robust emotional connections between customers and brands (Marketing Science Institute, 2016) and (2) improving emotional experiences (Temkin, 2018). Tackling the entirety of the CE elements simultaneously would require multiple methods tailored to each element and is beyond the scope of this paper.

The objective of this study is to develop and validate a novel and cost-effective approach, employing the recognized potential of AI in the service context to gain an improved holistic understanding of CE. Specifically, the study contributes to the CE literature in three key ways, via: (1) the development of a granular and semantically clear CE feeling model (CEFM) that is employable using AI techniques; (2) proposing a unique approach utilizing chatbots and SA to extract CE (feelings) from primary interview data and (3) empirically validating the effectiveness of this approach in extracting CE feelings. A detailed research agenda geared toward exploiting the potential of the proposed approach to inform and manage CE is outlined.

The following sections present an overview of the relevant CE and AI literature, details of the AI approach used, key findings and discussion of contributions and implications and concluding with limitations and a research agenda for future work.

2. Literature review

2.1 *The feeling component of customer experience*

There is a rapid growth in recognizing that “experience is everything”, acknowledging the importance of CE and that companies who focus their strategy on improving it could gain a competitive edge by bridging the experience gap (Clarke and Kinghorn, 2018). This relatively recent growth in the significance of CE in services has been evolved from the identification of related concepts including consumption experience (Holbrook and Hirschman, 1982) and the experience economy (Pine and Gilmore, 1998). Understanding and measuring experience effectively has been an ultimate goal for philosophers and researchers for some time, but it has proven difficult in practice.

CE is often defined as a dynamic and holistic, direct or indirect, interaction between a customer and a firm, involving elements of thoughts, feelings, activities, relations and sensations [1] (Lemon and Verhoef, 2016; Schmitt, 1999). The holistic nature of CE [2] renders this phenomenon challenging to observe and gauge accurately. Consequently, many practitioners tend to fall back on the use of dated tools such as Net Promoter Score (Reichheld, 2003) and CSAT (Anderson and Narus, 1984) as proxies (Maklan *et al.*, 2017). To address these shortcomings, tools such as the EXQ multiple-item scale, which leverages product experience, outcome focus, moments-of-truth and peace-of-mind (Klaus and Maklan, 2012), in addition to emerging text-mining approaches (e.g. McColl-Kennedy *et al.*, 2019) have been developed to evaluate CE. Nevertheless, as the importance and impact of (customer) experiences in our daily lives (Fisk *et al.*, 2020) keep increasing, further study of this subjective and holistic concept is warranted.

This study focuses on CE *feelings* specifically and responds to the surge of interest in examining this particular element of experience both in academia and management (Marketing Science Institute, 2016; McColl-Kennedy *et al.*, 2019). Drawing on a comprehensive review of the CE literature, the experiential element of feelings has been referred to as affect (e.g. Holbrook and Hirschman, 1982), feelings (e.g. Schmitt, 1999), emotions (e.g. Shaw and Ivens, 2002), hedonic value (e.g. Klaus and Maklan, 2011) or mood (Bagdare and Jain, 2013). Table 1 summarizes the semantic interpretations associated with the CE feeling element.

Accommodating the interpretations from Table 1 into a semantically precise model requires understanding how these terms differ in meaning and how they are represented within the fields of psychology concerned with studying subjective experience. Dividing these terms into two hierarchical levels (as shown in Table 1) allows for a natural grouping and increased semantic clarity. Provided that the hierarchical levels can be distinguished based on their temporal characteristics (Fox, 2018), it is argued here that feelings and affect act as higher-level umbrella terms (level 1) to the feeling element. Terms such as mood, emotion and hedonic value (level 2), on the other hand, represent a

Conceptual level	CE feeling term used	Key studies
1	Affect	Brakus <i>et al.</i> (2009), Csikszentmihalyi and Larson (1987)
	Feeling	Rajaobelina (2018), Klaus and Maklan (2011), Schmitt (1999)
2	Emotion	Heinonen (2018), Verhoef <i>et al.</i> (2009), Gentile <i>et al.</i> (2007)
	Hedonic value	Hilken <i>et al.</i> (2017), Verleye (2015), Klaus and Maklan (2011), Otto and Ritchie (1996)
	Mood	Bagdare and Jain (2013)

Note(s): Key studies (comprehensive review available from authors on request)

Table 1.
Consolidated
hierarchical
interpretations of CE
feelings

more experience-specific embodiment of feelings or affect (Babin *et al.*, 1994; Beedie *et al.*, 2005; Fox, 2018).

From the CE literature, it is also possible to find terms, such as entertainment (Ali *et al.*, 2016), leisure (Bagdare and Jain, 2013), safety (Otto and Ritchie, 1996) and joy (Bagdare and Jain, 2013), which all operate hierarchically below level 2. However, these level 3 concepts represent specific hedonic values attained within an experience and vary depending on the context under study (Babin *et al.*, 1994; Holbrook, 2006). Thus, to maintain robust model generalizability, CEFM that focuses on levels 1 and 2 is proposed.

The CEFM (Figure 1) represents the three sub-elements (at level 2; mood, emotion and hedonic value) reflecting their context and temporality characteristics (temporally prolonged, encounter-specific and context-specific) (Babin *et al.*, 1994; Fox, 2018). Capturing all three sub-elements simultaneously is vital for a holistic understanding of experiential feelings (Kranzbühler *et al.*, 2018). For instance, CE might be influenced by a customer engaging with a service in a negative mood. Similarly, emotions detail (one or more) encounter-specific feelings (i.e. toward firm resources and activities during the encounter) (Larivière *et al.*, 2017; Ordenes *et al.*, 2014), whereas hedonic value symbolizes how customers feel after having experienced the service. Combined, these sub-elements enable CEFM to compartmentalize different experiential feelings throughout the customer journey (pre, mid and post) (Lemon and Verhoef, 2016).

2.2 Enabling CE feeling extraction with AI

The purpose of the CEFM is to provide a consolidated, holistic and sufficiently granular framework for CE feelings to allow technologies such as AI to map experiential data onto it. AI is a prominent technological milestone that some claim ushered in the fourth industrial revolution (Maynard, 2015). AI technology, along with increasing levels of global (big) data production, generates new opportunities to advance and improve a myriad of research areas (Boyd and Crawford, 2012; Marr, 2018). Among these are service and technology (Mende *et al.*, 2019; Wirtz *et al.*, 2018), customer engagement (Kunz *et al.*, 2017), service encounters (Larivière *et al.*, 2017), sales (Syam and Sharma, 2018), customer relationship management (Berry and Linoff, 2004) and CE (Kabadayi *et al.*, 2019; Ordenes *et al.*, 2014; Zolkiewski *et al.*, 2017).

Companies must, however, understand the benefits and deficiencies of alternative approaches when they attempt to find the most appropriate AI technology. Each

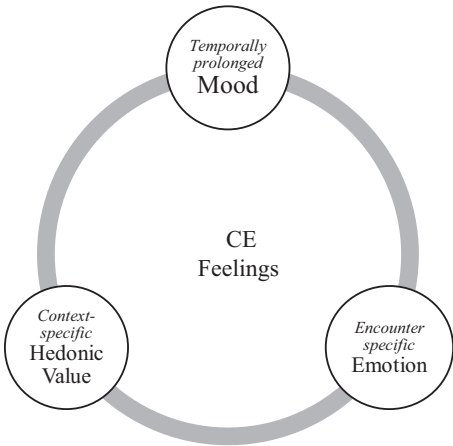


Figure 1.
Conceptual framework:
CE feeling
model (CEFM)

technology stack consists of various techniques, which utilize different AI methods to achieve a specific objective. For instance, supervised and unsupervised machine (deep) learning methods are useful for predicting results from either structured (e.g. tabular and transactional) or unstructured (e.g. captured images, audio, video or text) datasets (Brynjolfsson and McAfee, 2017). AI technologies involving data mining (e.g. text analysis), for example, add to the body of big data, which then feeds into machine learning algorithms and ultimately leads to improved prediction accuracy (Boyd and Crawford, 2012; Ng and Wakenshaw, 2017). In a storytelling environment based on experiential exchange via narrative, as in this study, an AI technology capable of extracting insights from unstructured conversations is required. This technology would need to analyze text via natural language processing to decipher not only feeling-related content but also context and temporality (Ordenes and Zhang, 2019).

Tackling this task is challenging, specifically for advanced AI methods that rely on secondary data sources (e.g. social media text) due to the possibility of incomplete data (i.e. data which might not be there) and passive data collection (i.e. there is no reciprocal data exchange). Therefore, primary data collection efforts are deemed necessary.

2.3 Rationale for using chatbot interviewers and sentiment analysis

When companies consider their (primary) data collection techniques, they are faced with several alternative methods. Each method possesses its unique set of strengths and weaknesses. Finding an appropriate balance between data quality and cost-effectiveness is the key. On the one hand, traditional (face-to-face) in-depth interviews represent a more engaging and personal method in which the interviewer can probe interviewees for clarification or more details (laddering). This method also provides critical supplementary data, such as facial gestures and body language (Novick, 2008). Surveys, on the other hand, are generally less costly and can be deployed faster and with more flexibility (not limited to time or location) while reaching a broader audience. When implemented using technology (e.g. online), surveys can provide realtime analysis of the results (Benoit *et al.*, 2017).

Chatbots can reap the benefits of both surveys and interviews. They already interact with customers in various service sectors such as banking and insurance and provide a cost-effective solution targeting a broad customer base independent of time and location (Riikinen *et al.*, 2018). However, by switching the more traditional chatbot role from acting as an information source (i.e. passively answering questions) to a more inquisitive role (i.e. proactively asking interview questions), the chatbot starts to resemble an interviewer and assimilates many of the advantages of widely used interview and survey methods. Thus, chatbot interviews have the potential to become an efficient and widely used AI approach capable of collecting primary data via conducting storytelling narrative interviews that are well suited to examine subjective constructs like CE.

In addition, chatbot technology can also handle multifformat data (i.e. text, audio/voice), support automation (e.g. automatic transcription and translation) and could be developed to seek required information via laddering and probing questions. It can also become more engaging by adapting its “personality” to the interviewee (i.e. it can assume a customizable persona based on current or past conversations) featuring; eye, electrodermal activity and facial gesture, tracking technologies widely considered to be useful for studying emotions (De Keyser *et al.*, 2019; Ng and Wakenshaw, 2017).

A comparison of key CE data collection approaches (traditional interviews, surveys and chatbot interviews), including their potential advantages, is depicted in Table 2. From this, the potential of chatbot interviews as an effective and efficient data collection approach of CE can be observed.

Table 2.
Comparison of key
CE data collection
approaches

Advantages	Traditional interviews	Surveys	Chatbot interviews
Rich data	X		X
Personal/empathetic	X		X
Engaging	X		X
Laddering and probing questions	X		A
Body language observation	X		A
Low cost		X	X
Broad reach/scalability		X	X
Fast deployment/speed		X	X
Flexible availability		X	X
Realtime analysis		X	X
Multiformat conversation availability			X
Automation			X
Adaptable personality			A
Note(s): “A” denotes further development potential via augmentation			

The extant literature recognizes that data mining – a set of tools and techniques utilizing statistics, AI and machines learning methods in order to discover meaningful patterns and rules in a dataset – has much potential for analyzing the data collected during a chatbot interview (Berry and Linoff, 2004, p. 7; Rygielski *et al.*, 2002). Since chatbots can leverage data-mining techniques to extract meaning from text (Feldman and Sanger, 2006; Ordenes and Zhang, 2019), they are thus technically capable of extracting CE feelings from collected text responses. This ability further adds to the value of our proposed approach in utilizing chatbots for data collection purposes.

More specifically, SA, a text-mining method used to determine the overall attitudes, opinions and emotions within text (Humphreys and Wang, 2018), could be used to map experiential interview feeling data to the CEFM. For instance, a question relating to a participant’s mood could be analyzed using SA, resulting in a polarity score for that statement (i.e. positive, negative and neutral). A typical SA algorithm would compare sets of terms from the input text against a sentiment lexicon such as WordNet, then calculate the distance between these terms and provide a polarity score as an output (Feldman, 2013). Thus, SA has the potential to attribute sentiment scores to experiential interview data, enabling companies to gain useful quantitative CE insights from natural language.

3. Method

In an attempt to improve the understanding of CE, this research adopts a bottom-up abductive approach using technologies that collect and evaluate experiential CE element data to build up a general and context-specific notion of the CE feeling concept (Brodie *et al.*, 2017). Developing a one-size-fits-all technological solution to the entirety of CE is not desirable, as studying different elements calls for different methodological strategies (e.g. SA works well for studying feelings but less so for studying sensations, where electrodermal activity measures and others may be more suitable). Focusing on a single CE element is also meaningful due to the fragmented literature on CE and its elements (Kranzbühler *et al.*, 2018; Palmer, 2010); high levels of semantic clarity are thus required for the effective implementation (e.g. extraction of experiential data) of the proposed approach.

Methodologically, this study consists of four inter-connected phases: data collection and preprocessing, scale validation, SA and the comparison between scale and sentiment scores. The phases are elaborated below.

3.1 Phase 1: data collection and preprocessing

In Phase 1, data were collected via a crowdsourcing platform, followed by data preparation, cleaning and filtering. Crowdsourcing platforms are satisfactory ways to provide a random pool of participants across which studies related to attitudes and behaviors can be conducted (Hulland and Miller, 2018). The platform employed in this study is Prolific Academic, which draws on a more transparent and diverse population of consumers more naïve to everyday experimental research tasks compared to platforms such as Amazon's MTurk (Palan and Schitter, 2018; Peer *et al.*, 2017).

Using Prolific Academic, 200 participants were recruited to interact with a chatbot named *Marvino*. Marvino was designed to ask interview-like questions about a recent service experience with a firm of the participants' choice. The chatbot interviewer first asked the participants to provide a descriptive, free-text answer to an inquiry, which was immediately followed by closed-end survey-type question(s) in the same (chatbot) conversation so that an understanding of mood, emotion and hedonic value was gained for comparison purposes. For example, questions on a participant's (recalled) mood would start by asking for a free-form textual response, followed by questions (items) adopted from an established and validated measurement scale (Table A2). Between each question, Marvino utilized conversational acknowledgments (e.g. "Thank you", "I see") to promote a more natural storytelling experience.

In order to enhance the generalizability of the CE findings, each participant was randomly allocated a personal experience to recall, consisting from four options. Positive and negative shopping and positive and negative vacation experiences ($N = 48$, $N = 53$, $N = 47$ and $N = 45$, respectively). Next, participants were asked how well they recalled the self-chosen service experience and were filtered out if their answer was less than 5 (i.e. "neutral") on the 9-point scale used. This left 193 participants in the final sample where a wide range of demographics are represented (Table A1).

IBM SPSS Statistics and AMOS 25 (2017) were used to analyze the resultant data. A missing data analysis revealed that only 2.1% of the data were missing. Little's missing completely at random (MCAR) test (Little, 1988) yielded a non-significant result ($\chi^2 = 27.745$, $df = 24$, $p = 0.271$) suggesting that the absent values can be assumed to be missing entirely by chance [3]. The multiple imputation (Rubin, 1987) approach was used, and a fully conditional specification method ($m = 10$) in SPSS IBM Corp (2017) was employed for estimating the missing values.

3.2 Phase 2: scale validation

In Phase 2, the reliability and validity of the measurement scales were tested from the data related to mood, emotion and hedonic value, as recalled by the participants. The procedure consisted of a maximum likelihood confirmatory factor analysis (CFA) conducted in AMOS 25 (2017). The measurement scales, along with the corresponding measurement items and their standardized loadings, relating to each feeling sub-element are shown in Table A2.

3.3 Phase 3: sentiment analysis and model performance metrics

In Phase 3, the effectiveness of the SA algorithm on the extracted CEFM data was validated. While there are many SA algorithms to consider, the Valence Aware Dictionary for Sentiment Reasoning (VADER) algorithm was adopted for its efficiency and performance at handling short and informal social media text (Hutto and Gilbert, 2014; Ribeiro *et al.*, 2016). Also, because VADER "was created from a generalizable, valence-based, human-curated gold standard sentiment lexicon", it exhibits the flexibility needed to handle human conversation where domain-specificity would be hard to determine (Ribeiro *et al.*, 2016, p. 7).

The performance of this algorithm is evidenced by how well it managed to predict and score each descriptive question of the CEFM. Specifically, the scores outputted by the algorithm were compared to their manually coded labeled counterparts. For finer sentiment granularity (Ordenes *et al.*, 2017), the compound scores (−1,1) produced by VADER were normalized to match the scale items in Table A2 (i.e. sentiment score of −1 refers to the scale score of 1 “extremely negative”; sentiment score of zero refers to 5 “neutral”; and sentiment score of 1 refers to 9 “extremely positive”). The statements then underwent a binary classification where they were split into (1) positive (scores ≥ 5) versus negative (scores < 5) cases, in line with the compound sentiment score provided by the SA algorithm and (2) true versus false, depending on whether the algorithm correctly predicted the statement sentiment (elaborated below).

To achieve this, a manual coding procedure involved the coder to first categorize the participant’s statement into one of the five categories (i.e. “extremely negative”, “average negative”, “neutral”, “average positive” and “extremely positive”) corresponding to the respective scale points of 1, 3, 5, 7 and 9. If the score of the SA algorithm fell within two scale points of the coder’s selected statement category (e.g. if the coder scores the statement as “average positive” and the sentiment score is 8, which is only 1 point from 7), then the coder would denote the prediction as true. Otherwise, the prediction was denoted as false. The potential reasons for false predictions were analyzed further to understand whether an inaccuracy was due to the SA algorithm or the participant; this and the inter-coder reliability are discussed in the results section.

Based on the automated (VADER) SA as well as the manual coding, each statement was categorized into a confusion matrix (Sokolova and Lapalme, 2009), splitting the results into true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) depending on their respective polarity (i.e. positive or negative classification by the algorithm and coders). This classification allowed for the calculation of key metrics (Table 4) that give valuable information about the effectiveness of the VADER SA algorithm. For a detailed account on these measures, refer to Sokolova and Lapalme (2009, p. 430).

4. Results

4.1 Phase 2: Scale validation results

The reliability and validity analysis, along with the descriptive statistics, of the key constructs under study are reported in Table 3. The final CFA model fitted the data well ($\chi^2 = 49.51$, $df = 25$, $p = 0.001$, $RMSEA = 0.07$, $SRMR = 0.02$, $GFI = 0.95$, $CFI = 0.99$, $TLI = 0.99$) (Brown, 2015; Hu and Bentler, 1999; Kline, 2016). In support of convergent validity, all relevant construct reliabilities (CR) were above the recommended level of 0.70 (Fornell and Larcker, 1981). The discriminant validity of the scales was also satisfactory as the relevant maximum shared variance (MSV) indices were shown to be greater than the corresponding average variance extracted (AVE) indices (Hair *et al.*, 2010), and the square-roots of the AVE appeared to be greater than the inter-construct correlations (Fornell and Larcker, 1981) as shown in Table 3.

Table 3.
Descriptive statistics,
correlations and
construct reliability
and validity

Construct	Mean (S.D.)	CR	AVE	MSV	1	2	3
1. Experience emotion ^a	5.29 (2.63)	–	–	–	–		
2. Experience mood	5.61 (2.42)	0.95	0.81	0.28	0.53	0.90	
3. Experience hedonic value	6.01 (2.66)	0.98	0.93	0.84	0.92	0.52	0.97
Note(s): Square-root of AVE on the diagonal in italic; correlations off-diagonal; ^a This construct consists of one item only; hence, CR, AVE and MSV are not meaningful							

	Pos	Neg	Total	Accuracy	Precision	Recall	F ₁ -score	Specificity	AUC	CE assessment via chatbot interviews
<i>Mood</i>										
True	110	60	170							
False	4	2	6							
Total	114	62	176	0.97	0.97	0.98	0.97	0.94	0.96	
<i>Emotion</i>										
True	104	58	162							
False	5	1	6							
Total	109	59	168	0.96	0.95	0.99	0.97	0.92	0.96	
<i>Hedonic value</i>										
True	103	52	155							
False	10	0	10							
Total	113	52	165	0.94	0.91	1.00	0.95	0.84	0.92	
<i>CEFM</i>										
True	317	170	487							
False	19	3	22							
Total	336	173	502	0.96	0.94	0.99	0.97	0.89	0.95	
Note(s): The F_1 harmonic mean was used as the F -score measure; CEFM refers to the customer experience feeling model, which consists of mood, emotion and hedonic value										

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Table 4.

The SA algorithm (VADER) performance on CEFM

4.2 Phase 3: Sentiment analysis and model performance metrics results

To determine the effectiveness of the VADER algorithm in correctly extracting the sentiments of the CE feelings in the dataset, a confusion matrix for each of the sub-elements was constructed. The inter-coder reliability kappa (Fleiss, 1971) of 0.95, for 60 [4] data-points for three coders across each of the CE feeling sub-elements, refers to an “almost perfect” agreement (Landis and Koch, 1977, p. 165).

The performance of the SA algorithm for each CE feeling sub-element is portrayed in Table 4. The results suggest that VADER is highly effective in extracting the sub-elements and CE feelings overall. Specifically, high accuracy, precision, recall, F_1 -score, specificity and AUC scores are reported (Sokolova and Lapalme, 2009). The lowest relative scores are reported for the extraction of hedonic value, whereas the highest effectiveness is evident for mood.

4.3 Phase 4: Comparison between scale and sentiment scores

While the findings in Phase 3 established that, in principle, the SA algorithm used is effective for studying CE feelings, the final phase (Phase 4) set out to address two important follow-up questions: (1) How closely the extracted sentiment scores from the AI chatbot interview match those of the respective scale scores and (2) to what extent the approach’s potential inaccuracies are due to the SA algorithm? Addressing these questions provides further understanding regarding the performance and future development potential of this approach.

In addressing the first question, the full sample ($N = 193$) was used to examine the similarities between the sentiment scores and the average measurement scale scores for each of the feeling sub-elements. The findings are promising, as the similarity percentages are 86.88% for mood, 84.66% for emotion and 84.33% for hedonic value. A series of paired t -tests further revealed that, at the 95% confidence level, the differences between the sentiment and scale scores are statistically insignificant for mood ($p = 0.18$) and emotion ($p = 0.07$), but statistically significant for hedonic value ($p = 0.03$).

To address the second question, post-hoc analyses with two sub-samples of the full dataset were conducted. Cases were included in these sub-samples based on the absolute percentage difference between the sentiment and scale scores and whether the substantial differences were considered to be due to the participants (i.e. input error) or the SA algorithm. Two explicit thresholds (18.75 and 12.5%[\[5\]](#)) for the absolute differences were chosen; the sub-samples thus consisted of all cases at or above (e.g. difference $\geq$ 12.5%) each threshold, where errors caused by the SA algorithm were included, but input errors caused by the participants were not.

Therefore, the two sub-samples were manually coded to distinguish between input errors (i.e. participant-induced errors such as spelling and unclear statements) and errors caused by the SA algorithm (VADER). The coding scheme, along with code descriptions and illustrative examples from our dataset, is presented in [Table A3](#). The inter-coder reliability for two independent coders resulted in a kappa ([Fleiss, 1971](#)) of 0.62 for the 259 data-points across the CE feeling sub-elements, suggesting “substantial agreement” ([Landis and Koch, 1977](#), p. 165).

The distribution of the coded items for both thresholds (i.e. difference $\geq$ 18.75 and 12.5%) are shown in [Table 5](#). The number of cases identified at the 18.75% threshold varies between 43 and 61 and expectedly increases to range between 82 and 97 at the 12.5% threshold. Evidently, “Statement-Scale Mismatch” (at the 12.5%: 60.98% for mood, 56.10% for emotion and 43.30% for hedonic value) represents the most prominent reason for the differences in the sentiment and scale scores, followed by the sentiment algorithm accuracy, which at the 12.5% threshold explains between 19.51% (mood) and 31.96% (hedonic value) of the discrepancies.

A series of paired *t*-tests, reported in [Table 6](#), concludes the post-hoc analyses. In the two sub-samples, all such cases where participant-induced errors are the underlying cause for the difference in sentiment and scale scores are excluded. The results suggest that isolating the effect of sentiment algorithm accuracy improves the similarity percentages for all feeling sub-elements. Specifically, using the tighter threshold (12.5%), the similarities improve to 92.58% (mood, *N* = 127), 89.90% (emotion, *N* = 128) and 90.27% (hedonic value, *N* = 127). The mean differences also decrease as the thresholds are applied and, more importantly, the hedonic value’s score differences become statistically insignificant (*p* = 0.06). In sum, the findings in [Table 6](#) show that the two approaches produce closely similar scores, especially when input errors are accounted for. The potential improvements to the approach proposed in this study are discussed in [Section 5.3](#).

Table 5.
Distribution of the
coded items

Sub-elements/codes	Experience mood		Experience emotion		Experience hedonic value	
	18.75%	12.5%	18.75%	12.5%	18.75%	12.5%
Threshold	18.75%	12.5%	18.75%	12.5%	18.75%	12.5%
Sentiment algorithm accuracy	13.95%	19.51%	21.05%	20.73%	22.95%	31.96%
Statement unclear	2.33%	3.66%	5.26%	6.10%	8.20%	6.19%
Statement language	6.98%	3.66%	3.51%	2.44%	3.28%	2.06%
Statement irrelevant	0%	0%	8.77%	8.54%	16.39%	10.31%
Statement-scale mismatch	65.12%	60.98%	56.14%	56.10%	44.26%	43.30%
Statement multiple context/time	11.63%	12.20%	5.26%	6.10%	4.92%	6.19%
Total items	43	82	57	82	61	97
Note(s): Denotes how many items out of the full sample (<i>N</i> = 193) were coded at each threshold						

CE feeling sub-element	Sample ^a	Similarity % ^b	Mean difference (S.D.)	Std. Error	<i>t</i> (df)	Sig. (2-Tailed)	CE assessment via chatbot interviews
Experience mood	Full sample	86.88%	0.14 (1.43)	0.10	1.36 (192)	0.18	755
	Threshold 18.75%	91.25%	0.06 (0.87)	0.07	0.87 (155)	0.39	
	Threshold 12.5%	92.58%	0.04 (0.76)	0.07	0.56 (126)	0.58	
	Full sample	84.66%	0.21 (1.61)	0.12	1.82 (192)	0.07	
Experience emotion	Threshold 18.75%	89.23%	0.06 (1.16)	0.10	0.63 (147)	0.53	
	Threshold 12.5%	89.90%	0.03 (1.16)	0.10	0.26 (127)	0.80	
	Full sample	84.33%	0.24 (1.56)	0.11	2.16 (192)	0.03	
	Threshold 18.75%	89.51%	0.23 (0.97)	0.08	2.84 (145)	0.01	
Experience hedonic value	Threshold 12.5%	90.27%	0.16 (0.94)	0.08	1.88 (126)	0.06	
	Full sample	86.88%	0.14 (1.43)	0.10	1.36 (192)	0.18	
	Threshold 18.75%	91.25%	0.06 (0.87)	0.07	0.87 (155)	0.39	
	Threshold 12.5%	92.58%	0.04 (0.76)	0.07	0.56 (126)	0.58	

Note(s): ^a The “Threshold 18.75%” and “Threshold 12.5%” sub-samples include all cases at or above the difference threshold where the differences were caused by the SA algorithm (i.e. excludes all cases with participant’s input errors); ^b “Similarity %” refers to the percentage similarity between the sentiment score and the average scale score

Table 6. Similarity percentages and paired difference *t*-tests for the feeling sub-elements

5. Discussion

The objective of this study was to develop and validate a novel and cost-effective approach, drawing on the recognized potential of AI in the service context for gaining an improved understanding of CE. The empirical findings support the effectiveness of SA chatbot interviews for eliciting and examining CE feelings, highlighting the promise that the approach holds in the digital age. The key theoretical and methodological contributions, practical implications, research agenda and limitations are highlighted and discussed below.

5.1 Theoretical and methodological contributions

This study contributes to the service management literature in three main aspects. First, a granular and semantically clear framework (CEFM) for studying CE feelings is developed via a comprehensive literature review. The CEFM informs CE theory by providing a holistic understanding of CE feelings, consisting of three complementary sub-elements: mood (temporally prolonged feelings) (e.g. Beedie *et al.*, 2005), emotion (encounter-specific feelings) (e.g. Fox, 2018) and hedonic value (context-specific feelings) (e.g. Babin *et al.*, 1994). In line with the recent service literature, the framework recognizes that feelings need to be treated as a continuous phenomenon occurring within and outside of the service context, where the experience of service encounters is linked with broader human experience (Fisk *et al.*, 2020; Lemon and Verhoef, 2016). The semantic clarity embedded into CEFM is critical, as this makes it deployable by AI techniques such as text mining (Ordenes *et al.*, 2017). Developing such semantic clarity within the experience domain of service research also responds to the recent call (Fisk *et al.*, 2020) for researchers to develop service language that can be used to facilitate improved wellbeing for everyone. Specifically, the CEFM distinguishes the different feeling sub-elements and facilitates the understanding of CE feelings holistically.

Second, leveraging the CEFM, a novel approach where a chatbot and SA are employed to collect primary data and extract valuable CE (feeling) insights is proposed. The approach offers distinct advantages over alternatives as it addresses notable weaknesses of traditional interviews (e.g. high costs and lack of scalability) and survey approaches (e.g. lack of engagement, limited depth of data and poor response rates). Using a chatbot as an interviewer

would enable service companies to collect rich primary data from large samples quickly and cost-effectively. This fundamentally atypical use of chatbots (i.e. proactive rather than reactive) helps alleviate many of the potential downsides of traditional interviews since it inherits the efficiencies typically related to surveys (Benoit *et al.*, 2017). Using an AI-augmented chatbot interviewer to analyze and (immediately) respond to CE (feelings) can also improve the way CE is managed.

An AI-enriched chatbot interview approach possesses two important characteristics: immersion and engagement. Compared to a survey, Marvinno adds the crucial element of a reciprocal – even if simple and linear – immersive conversational narrative around experiences. Some participants highlighted that they felt being part of a conversation that helped them express their “experience story” via the “chat experience” (participant 85). Engagement, on the other hand, helps establish personal trust and gratitude: “It makes the exchange more personal and adds a bit of fun/imagination” (participant 193). Naming the chatbot Marvinno created an approachable human-like anthropomorphic persona, further signifying the natural human tendency to exchange experience via storytelling: “the conversation was very nice and this study was an interesting experience for me;-) Goodbye Marvinno! Thanks for this conversation;-)” (participant 172); “it was comfortable to chat with you, Marvinno” (participant 133); and more playfully, “you’re perfect Marvinno do not let no-one tell you different . . . you should become a therapist” (participant 54). The above comments show how valuable natural storytelling narratives can be; by helping customers relive and surface their experiences through conversational retelling, more profound holistic inputs and outputs can be achieved with the use of AI-augmented chatbot interviewers.

Third, the key findings demonstrate that the SA algorithm used in this study (VADER) was able to extract the sentiments expressed by the participants adequately. It did particularly well with mood and emotion, struggling slightly more with hedonic value; this might be due to the latter indirectly portraying an outcome feeling (i.e. how a participant feels as a result of a service encounter) as opposed to an explicit feeling (i.e. those of mood and emotion). Direct comparisons between the extracted sentiment scores and the measurement scale scores offered further validation for the proposed method; the two approaches produce closely similar scores especially when input errors are accounted for. The post-hoc analyses reveal that it is essential to distinguish between the SA algorithm being at fault (i.e. when the sentiment scores do not match the text description) and participant-induced errors (e.g. typos). In sum, this study makes a convincing case for the promise that the proposed approach holds. Thus, developing AI-enabled frameworks such as CEFM, coupled with an analytical chatbot implementation, appears effective for measuring aspects of CE and consequently complements existing methods such as EXQ and generic text mining. This is achieved by incorporating further qualitative analysis of the raw output conversations (Kuppelwieser and Klaus, 2020). The findings also point to the need for further refinements to this approach; how this could be done in practice is discussed below.

5.2 Practical implications

“I’ve never spoken to a chatbot before! I must say, I found it interesting and futuristic! I love it! (Respondent 18)”

Each of the key contributions of the study also points to potential managerial implications. First, adopting and applying the CEFM logic would allow service companies to establish a deeper understanding of individual CE feelings, serving as a useful guide for any adjustments to service experience design. Importantly, managers could not only develop an understanding of how individual service encounters affect customer feelings, but also how their mood or extrinsic influences affect their hedonic perception of the service overall in the context of their wider human experience (Fisk *et al.*, 2020).

Second, the use of conversational agents for narrative inquiry has considerable potential for positive business impact. They serve as cost-effective tools to collect primary data on a large scale. A fundamental advantage of using chatbot interviewers lies in the manner in which they direct the conversation and ask participants explicit questions that help unpack complex phenomena such as CE. Without a chatbot asking for specific sentiments related to the feeling sub-elements, the adoption of an analytical AI approach relying on secondary data only, would need to be more complex, use massive training datasets and assume that the data it requires is correctly labeled and readily present for analysis. Thus, by using chatbots to collect primary data, there is less guesswork involved, especially as chatbots can simply ask a question again if they fail to understand the answer, much like a human interviewer would. Moreover, the approach facilitates the extraction of feelings that might not be defined in a discrete pre-defined spectrum (e.g. anger, fear, joy and surprise), thus complementing such approaches (Madhala *et al.*, 2018). Thus, by providing chatbots with a positive persona and demonstrated empathetic reciprocal feelings within the interview experience, these conversational agents become useful tools enabling companies to foster immersive and engaging relationships with customers, resulting in improved CE.

Third, the empirically validated novel AI-ready tool, combining chatbot-enabled data collection and text analysis via SA, offers a cost-effective approach for companies to enhance their understanding of CE. The customer-centric approach helps service companies to bridge the current gap in CE measurement enabling a resource-effective analytical implementation. Moreover, chatbot interviewer technology comes with considerable versatility as it offers companies the potential to collect and analyze diverse (e.g. in terms of demographics) and complementary qualitative and quantitative datasets from multiple channels (e.g. live chat and company database). For example, chatbots could tap into local news outlet services and adapt their personality to the macro context's overall sentiment. This could provide a more genuine storytelling experience (e.g. an airline chatbot could sense an overall negative sentiment due to weather-related cancellations and could portray more empathetic responses as a result), which in turn could add to the insights gained during conversation.

5.3 Future research agenda

Drawing on the above discussion, Table 7 outlines a detailed future research agenda, which consists of research topics as well as managerial and technical concerns for the CEFM, the chatbot interviewer and the SA algorithm. Further research topics for each of these are outlined within the contexts of CE feelings, CE elements and service management.

Concerning the CEFM, the framework depicted in this study assumes that the sub-elements are related but static. However, in line with Kranzbühler *et al.* (2018), it is posited that a more dynamic structure could enable the measurement of CE feelings even more effectively. Development of such a dynamic model could, for example, take into consideration the temporality of feelings (e.g. mood before and after service encounter) and multiple emotions exhibited during multiple encounters and thus help paint a more detailed and holistic picture of the customer's feelings. How CE feelings and its sub-elements affect other CE elements (e.g. CE thoughts) and service management more broadly would also need to be investigated further.

A relevant question calling for further examination pertains to reducing problematic participant-induced errors (e.g. unclear statement, statement mixes multiple contexts, statement irrelevant) in the chatbot interviews. A potential avenue to address this in the multiple context scenario could, for example, be the incorporation of realtime target-based SA, which would trigger the chatbot to ask for further clarifications to single out the different contexts (Carvalho *et al.*, 2009). Additionally, future research should examine to what extent and how chatbot interviews might influence the participant's experience and the data collected. It would also be useful to identify other service management research areas that

Context	Research topics	Managerial/technical concerns
<i>CE feeling model (CEFM)</i>		
CE feelings	• How does temporality affect each of the feeling sub-elements? (e.g. how mood before and after an encounter influences overall CE feelings)?	• How could managers score (attribute weights to) different feeling sub-elements?
CE elements	• How do the feeling sub-elements influence other CE elements?	• How do feeling sub-elements influence CE decisions compare to other CE elements?
Service management	• How do feeling sub-elements influence different service constructs (e.g. customer engagement and transformative service research)?	• How do feeling sub-elements influence different service management facets (e.g. service design, customer journey)?
<i>Chatbot interviewer</i>		
CE feelings	• What is the influence of a chatbot interviewer on the level of expression of participant CE feelings?	• What are possible methods for discerning whether participants expressed valid answers to the chatbot's questions?
CE elements	• How well can a chatbot interviewer address and extract the entire CE element spectrum?	• What strategies, incentives and levels of effort would be required for participants to adequately provide the details of their CEs?
Service management	• How can different service management research areas benefit from a chatbot interviewer?	• How would employment of chatbot interviewers in service management impact costs and resource utilization?
<i>Sentiment analysis</i>		
CE feelings	• How would different algorithms fair against VADER, and how could SA be improved?	• What technical improvements could be made to improve the performance of SA of CE feelings?
CE elements	• To what extent is SA an effective method of extracting the remaining CE elements?	• What technical SA algorithm modifications would be necessary when evaluating different CE elements?
Service management	• How can SA be used to further service constructs (e.g. service failure)?	• What are the limitations of the current application of SA in different service contexts? How could these limitations be overcome?

Table 7.
Research agenda

could benefit from the deployment of a chatbot interviewer, while examining the key managerial and technical challenges involved in incorporating this approach.

Finally, the preprocessing phase of the SA algorithm, which is often portrayed in data-mining frameworks such as CRISP-DM, could be further optimized (Azevedo and Santos, 2008). Although the implementation of this phase differs between solutions, for this particular case, removing a more extensive array of stop words and non-influential parts-of-speech (Ordenes and Zhang, 2019), including spelling and grammar checking could precede the SA task. This would result in potentially improving CE feeling extraction accuracy, as highlighted in the managerial/technical SA portion of Table 7.

5.4 Limitations

Despite the promising empirical results reported in this study, some limitations naturally need to be noted. First, the chatbot utilized here was a non-responsive conversational agent. This forced participants to restart the entire conversation if they wanted to modify a previously inputted answer, and this could have affected the results if several participants

chose not to reinitiate the conversation again. Using a more advanced chatbot with the option to conversationally request to track back and change a previous answer (as you can in natural discourse) or to clarify a point (as discussed previously) would address this limitation. Second, the fact that European participants (82.9%) represent an overwhelming majority in the sample, which is also skewed toward the younger participants (69.4% of the sample are below 35 years of age), could also affect the findings and needs to be considered when generalizing the findings to a broader population [6]. Third, the study did not account for technology biases the participants could have exhibited and how these would impact the interaction and recollection with the chatbot. For instance, participants might be biased against human or conversational agents. Researchers could address this by incorporating questions that assess potential technology bias. Fourth, to preserve storytelling/narrative immersion, a simple affect model was utilized to measure emotions to alleviate the technical and lengthy measurement scales of more complex variations – such as Mehrabian and Russell's model (1974) pleasure, arousal, dominance, or Plutchik's (1980) emotional model – even if these might have resulted in a slightly more accurate understanding of the sub-elements of CE. Fifth, as noted, a small number of missing values appeared as a result of connectivity (i.e. network) issues between chatbot and participants, thus researchers should consider Internet stability in their implementations of this approach. Lastly, VADER was the only SA algorithm used due to the primary objective of the study, which aims at understanding how well SA fairs against validated measurement scales of each of the feeling sub-elements. Testing how other SA algorithms fair against VADER would be interesting and would shed more light on the effectiveness of the proposed chatbot approach. This, as well as the other limitations, could be addressed in future work.

Notes

1. While other elements, such as the economic and lifestyle elements (Gentile *et al.*, 2007; Verleye, 2015) have also been proposed, the above five elements are the most widely adopted (De Keyser *et al.*, 2020).
2. Experience has been conceptualized as being “life itself” (Csikszentmihalyi, 1991, p. 192)
3. Upon further analysis, the missing values appear to be a result of connectivity (i.e. network) issues between the chatbot and the participants.
4. A full coding assignment was not deemed necessary, since VADER is already a validated SA algorithm (Hutto and Gilbert, 2014)
5. These, respectively, refer to ± 1.5 and ± 1 scale point difference on the 9-point scale used
6. Post-hoc tests revealed that our key findings are statistically indifferent for European (vs others) and under 35 year olds (vs others), lending support for wider generalizability.

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Appendices

Demographic (N = 193)		Frequency	Percent %
Age	18–24	72	37.3
	25–34	62	32.1
	35–44	29	15.0
	45–54	20	10.4
	55+	10	5.2
Gender	Female	87	45.1
	Male	101	52.4
Region	Undisclosed	5	2.5
	Australia	1	0.5
	Balkans	9	4.7
	Europe	160	82.9
	Middle East	3	1.6
	North America	12	6.2
	South America	1	0.5
	Southeast Asia	1	0.5
Mobile device	Undisclosed	6	3.1
	Yes	50	25.9
Student	No	143	74.1
	Yes	65	33.7
	No	123	63.7
	Undisclosed	5	2.6

Table A1.
Sample frequencies

				CE assessment via chatbot interviews
Source	Constructs	Chatbot questions ^a	Stand. Loading	
Mood short form (MSF) (Peterson and Sauber, 1983)	Experience mood	"You would say your mood in general THAT day was . . ." (Extremely negative – Extremely positive)	0.881	765
		"THAT day you recall feeling . . ." (Extremely Dull - Extremely Cheerful)	0.847	
		"How emotionally comfortable or uncomfortable do you remember feeling THAT day?" (Extremely Uncomfortable - Extremely Comfortable)	0.795	
		"As per your recollection, how did you feel THAT day?" (Extremely Tense - Extremely Calm)	0.729	
Hedonic consumer attitudes (Batra and Ahtola, 1991)	Experience hedonic value	"Overall this experience was . . ." (Extremely Displeasing - Extremely Nice)	0.970	Table A2. Measurement items and standardized loadings
		"In the end you felt the experience was . . ." (Extremely Unpleasant - Extremely Pleasant)	0.944	
		"How agreeable or disagreeable would you say the whole service experience was?" (Extremely Disagreeable - Extremely Agreeable)	0.896	
		"This experience left you feeling . . ." (Extremely Sad - Extremely Happy)	0.917	
Watson and Tellegen (1985)	Experience emotion	"How positive/negative was this interaction ^b ?" (Extremely Negative - Extremely Positive)	–	
Note(s): ^a All responses were obtained using 9-point scales; ^b "this interaction" refers to a context built by previous questions inquiring about a memorable service encounter moment, and the service resource (object or person) (Ordenes <i>et al.</i> , 2014)				

Code	Description	Examples	
Sentiment algorithm accuracy	The algorithm did not pick up sentiments accurately	"Upset as it was a special occasion" (sentiment score 5.1) "I did not feel happy because of the meal" (sentiment score 7.3)	Table A3. Phase 4 coding scheme
Statement unclear	The statement did not convey the intended sentiments correctly and was not clear	"Awkward", "I felt normal", "more secure"	
Statement language	Errors with grammar and spelling affecting sentiment words	"Annoid", "irritability", "10/10"	
Statement irrelevant	The participant did not provide (any) sentiments targeted at the sub-element in question	"[company x] flight staff", "Felt like my problem was important", "She was very nice and understanding"	
Statement-scale mismatch	A mismatch between the scales answered and the description (e.g. an average sentiment but an extreme scale response)	"very disappointed and angry" (avg. scale score 5.8) "I was very calm" (avg. scale score 9.0)	
Statement multiple context/ time	Multiple timelines/contexts in the description	"first I was a bit shocked, then sad, afterward I was angry", "I have been in a very good mood but feel a bit down this evening"	

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